

Lab 5

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Switch>enable
Switch#config t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#hostname s1
s1(config)#vlan 10
s1(config-vlan)#name FacultyStaff
s1(config-vlan)#exit
s1(config)#vlan 20
s1(config-vlan)#name Students
s1(config-vlan)#exit
s1(config)#vlan 30
s1(config-vlan)#name Guest (DEFAULT)
s1(config-vlan)#
^
% Invalid input detected at '^' marker.

s1(config-vlan)#name Guest
s1(config-vlan)#exit
s1(config)#vlan 99
s1(config-vlan)#name Management
s1(config-vlan)#exit
s1(config)#
s1(config)#
%LINK-5-CHANGED: Interface FastEthernet0/1, changed state to up

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Switch>enable
Switch#config t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#hostname s2
s2(config)#vlan 10
s2(config-vlan)#name FacultyStaff
s2(config-vlan)#exit
s2(config)#vlan 20
s2(config-vlan)#name Students
s2(config-vlan)#exit
s2(config)#vlan 30
s2(config-vlan)#name Guest
s2(config-vlan)#exit
s2(config)#vlan 99
s2(config-vlan)#name Management
s2(config-vlan)#exit
s2(config)#

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Switch>enable
Switch#config t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#hostname s3
s3(config)#vlan 10
s3(config-vlan)#name FacultyStaff
s3(config-vlan)#exit
s3(config)#vlan 20
s3(config-vlan)#name Students
s3(config-vlan)#exit
s3(config)#vlan 30
s3(config-vlan)#name Guest
s3(config-vlan)#exit
s3(config)#vlan 99
s3(config-vlan)#name Management
s3(config-vlan)#exit
s3(config)#

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s2>enable
s2#config t
Enter configuration commands, one per line. End with CNTL/Z.
s2(config)#interface f0/11
s2(config-if)#switchport mode access
s2(config-if)#switchport access vlan 10
s2(config-if)#exit
s2(config)#interface f0/18
s2(config-if)#switchport access vlan 20
s2(config-if)#exit
s2(config)#interface f0/6
s2(config-if)#switchport mode access
s2(config-if)#switchport access vlan 30
s2(config-if)#end
s2#
%SYS-5-CONFIG_I: Configured from console by console

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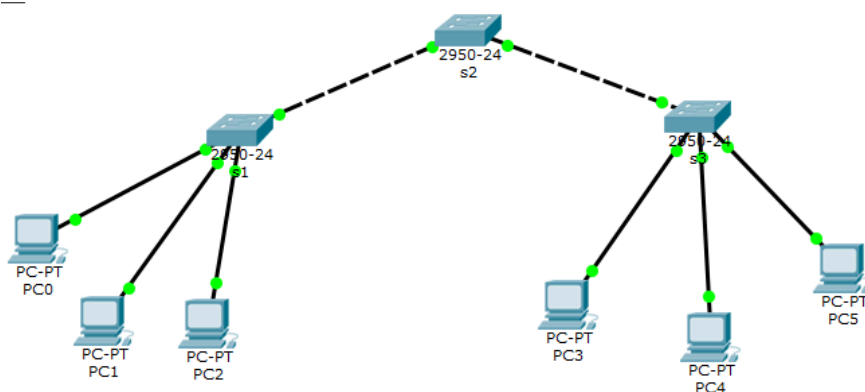
s3>enable
s3#config t
Enter configuration commands, one per line. End with CNTL/Z.
s3(config)#interface f0/11
s3(config-if)#switchport mode access
s3(config-if)#switchport access vlan 10
s3(config-if)#exit
s3(config)#f0/18
^
% Invalid input detected at '^' marker.

s3(config)#interface f0/18
s3(config-if)#switchport mode access
s3(config-if)#switchport access vlan 20
s3(config-if)#exit
s3(config)#interface f0/6
s3(config-if)#switchport mode access
s3(config-if)#switchport access vlan 30
s3(config-if)#end
s3#
%SYS-5-CONFIG_I: Configured from console by console

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s2#show vlan brief
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VLAN	Name	Status	Ports
1	default	active	Fa0/1, Fa0/2, Fa0/3, Fa0/4 Fa0/5, Fa0/7, Fa0/8, Fa0/9 Fa0/10, Fa0/12, Fa0/13, Fa0/14 Fa0/15, Fa0/16, Fa0/17, Fa0/19 Fa0/20, Fa0/21, Fa0/22, Fa0/23 Fa0/24
10	FacultyStaff	active	Fa0/11
20	Students	active	Fa0/18
30	Guest	active	Fa0/6
99	Management	active	
1002	fddi-default	active	
1003	token-ring-default	active	
1004	fddinet-default	active	
1005	trnet-default	active	



Which command will only display the VLAN name, status, and associated ports on a switch?

show vlan brief

What is the Advantage of using the VLANs in your Network?

VLANs provide a number of advantages including ease of administration, confinement of broadcast domains, reduced network traffic, and enforcement of security policies.

Ease of administration

VLANs enable logical grouping of end-stations that are physically dispersed on a network. When users on a VLAN move to a new physical location but continue to perform the same job function, the end-stations of those users do not need to be reconfigured. Similarly, if users change their job function, they need not physically move: changing the VLAN membership of the end-stations to that of the new team makes the users' end-stations local to the resources of the new team.

Reduction in network traffic

Confinement of broadcast domains on a network significantly reduces traffic.

Confinement of broadcast domains

VLANs reduce the need to have routers deployed on a network to contain broadcast traffic. Flooding of a packet is limited to the switch ports that belong to a VLAN.